Çankaya University
Computer Engineering Department
CENG 407 - Literature Review

AI-Based Dynamic Traffic Flow and Speed Management System

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Abstract

Peak-hour congestion on signal-free arterial roads is driven not only by high demand but also by unstable driving behaviour: repeated acceleration and braking amplifies small disturbances into backward-propagating stop-and-go waves that reduce effective capacity, increase travel time and raise fuel consumption. This literature review surveys the research our project builds on. We first summarise the traffic flow theory that explains shock waves and phantom jams. We then review Variable Speed Limit (VSL) and speed harmonization strategies, from rule-based and feedback controllers to model predictive control and (multi-agent) reinforcement learning, together with their field test results. Next, we review short-term traffic forecasting, from classical time-series models to spatio-temporal graph neural networks. We also cover vehicle-based approaches such as Lagrangian control and jam-absorption driving, studies and open data from Turkey, and the datasets and simulators used in the field. Finally, we compare similar deployed systems and identify the research gap that our project addresses: coupling a learned short-term congestion forecast with proactive, segment-level speed recommendations.

1 Introduction

Traffic congestion is commonly explained by the number of vehicles exceeding road capacity. Empirical and experimental research shows a second, equally important mechanism: when vehicle density passes a critical value, traffic flow becomes unstable, and small speed fluctuations grow into stop-and-go waves that travel upstream against the direction of traffic [1]. Once flow breaks down, the discharge rate from the queue is lower than the capacity before the breakdown, so the road effectively loses throughput exactly when it is needed most.

Variable Speed Limits (VSL), also called speed harmonization, aim to prevent or dissolve these waves by temporarily lowering and equalising speeds upstream of a disturbance. VSL systems have been operated in Europe for decades and have produced clear safety benefits, while their effect on traffic efficiency remains debated [2, 3]. At the same time, advances in deep learning have made accurate short-term traffic forecasting possible [4, 5], and reinforcement learning has emerged as a data-driven alternative for designing VSL controllers [6, 7].

Our project, the *AI-Based Dynamic Traffic Flow and Speed Management System*, combines these two directions. It divides a signal-free arterial road into segments, predicts short-term congestion risk from speed, density, flow and occupancy data, and converts these predictions

into safe and practical speed recommendations for each segment. This review is organised as follows: Section 2 covers traffic flow fundamentals, Section 3 reviews VSL control, Section 4 reviews short-term forecasting, Section 5 presents the proposed architecture, and the remaining sections cover models, datasets, deployment challenges, key papers, similar systems and our conclusions.

2 Traffic Flow Fundamentals: Shock Waves and Stop-and-Go Traffic

2.1 Macroscopic Models

The classical Lighthill–Whitham–Richards (LWR) model treats traffic as a compressible fluid described by flow q, density k and speed v, with $q = k\,v$ [8, 9]. When two traffic states meet, a shock wave forms whose speed is

$$v_{sw} = \frac{q_b - q_a}{k_b - k_a}, \tag{1}$$

where b and a denote the states before and after the change. A negative value means that the wave travels upstream, which is the typical behaviour of a congestion front. Discretised versions of this theory, such as the Cell Transmission Model and METANET, are the prediction models most often used inside VSL controllers [10].

2.2 Microscopic Models

Car-following models describe each vehicle's acceleration as a function of its speed, gap and the speed of the leading vehicle. The Intelligent Driver Model (IDM) [11] is widely used in simulators such as SUMO and is able to reproduce the formation of stop-and-go waves. Microscopic models are essential for our project, because the evaluation of speed recommendations requires simulating how individual drivers respond to them.

2.3 Phantom Jams and Capacity Drop

Sugiyama et al. placed 22 vehicles on a circular track and asked drivers to travel at a constant speed [1]. Although the road had no bottleneck, a jam cluster formed and propagated backwards, showing that a bottleneck is only a trigger and not the essential origin of a jam. This result motivates our project directly: if jams emerge from unstable flow, then detecting instability early and harmonising speeds can delay or prevent breakdown. For this reason we choose speed variance and stop-and-go frequency, not only average speed, as success metrics.

3 Variable Speed Limits and Speed Harmonization

VSL systems display dynamic speed limits on variable message signs. Two main functions are identified in the literature: speed homogenisation, which reduces speed differences between vehicles and lanes, and the prevention of traffic breakdown, which keeps the density at a bottleneck below its critical value [2]. A third concept, Mainstream Traffic Flow Control (MTFC), uses VSL upstream of a bottleneck as a "mainline metering" device to regulate the arriving flow [12].

3.1 Evidence on Effectiveness

Lu and Shladover reviewed simulation studies, field tests and combinations of VSL with ramp metering [2]. European practices generally reported benefits for both safety and flow, with safety being more significant, while United States practices showed clear safety improvements but controversial effects on traffic flow. Papageorgiou et al. analysed real data from a European

motorway and found no clear evidence of improved efficiency in operational VSL systems; however, VSL changed the flow–occupancy relationship, shifting the critical occupancy to higher values [3]. These findings suggest that our project should treat flow stability and safety as primary goals and throughput as a quantity to be maintained.

3.2 Rule-Based and Feedback Controllers

The SPECIALIST algorithm uses shock wave theory to decide where and how long to lower the speed limit in order to resolve a moving jam [13]. It was tested from September 2009 to February 2010 on a 14 km stretch of the Dutch A12 freeway [14]. When activated for shock waves, it resolved them in nearly 80% of cases; about half of the activations, however, occurred for other jam types, where effectiveness dropped to 40–50%. This shows that reliable detection of the jam type is as important as the control law itself. Carlson et al. compared optimal control and local feedback control for MTFC and found the feedback scheme suitable for immediate practical application, with low sensitivity to its parameters [12].

3.3 Model Predictive Control

Hegyi et al. formulated the coordination of VSL and ramp metering as a model predictive control (MPC) problem over a METANET model [10]. MPC optimises a cost such as total time spent over a rolling horizon and can include constraints on how fast speed limits may change. It is a natural choice for an "optimization-based" control layer, but its performance depends on the accuracy of the underlying traffic model.

3.4 Reinforcement Learning Approaches

Reinforcement learning (RL) learns a control policy by interacting with a simulated environment. Li et al. applied Q-learning to VSL at freeway recurrent bottlenecks [15]. Wu, Tan and Ran proposed a deep actor-critic controller for differential VSL, where different limits are set per lane, and compared reward signals based on total travel time, bottleneck speed, emergency braking and emissions [6]. Han et al. designed an RL-based VSL controller that targets freeway jam waves specifically [16]. Multi-agent RL (MARL) treats each controlled segment as an agent: Kušić et al. used distributed W-learning with four agents in SUMO, allowing agents to adjust the position and length of VSL zones as well as the limits [7]. In 2024, MARL-based VSL controllers were deployed in the field on Interstate 24 in the United States [17]. Since every road segment in our system can be modelled as an agent, a centralized-training, decentralized-execution (CTDE) MARL design is a promising extension of the rule-based baseline.

4 Short-Term Traffic Forecasting

Short-term traffic forecasting predicts speed, flow or travel time a few minutes to an hour ahead. Vlahogianni et al. reviewed the field and listed ten open challenges, including responsiveness to non-recurrent events and the integration of forecasts with control [4]. Early work used ARIMA and other statistical time-series models; later work introduced recurrent neural networks such as LSTM.

Because a road network is naturally a graph, graph neural networks (GNNs) have become the leading approach since 2018. Jiang and Luo surveyed more than 200 GNN-based traffic forecasting papers [5]. Representative models include DCRNN, which combines diffusion convolution with a recurrent encoder–decoder [18]; STGCN, which uses graph convolutions with gated temporal convolutions [19]; and Graph WaveNet, which learns an adaptive adjacency matrix [20]. The DL-Traff benchmark provides open implementations of these models on standard datasets [21].

Our proposal frames congestion prediction partly as a classification problem (evaluated with F1), while most of the literature performs regression on speed or flow (evaluated with MAE, RMSE and MAPE). We will therefore predict segment speeds and derive a congestion-risk label by thresholding, so that results can be reported with both groups of metrics.

5 Proposed System Architecture

Figure 1 shows the pipeline described in our proposal. Raw traffic data is cleaned and turned into features such as average speed, density, flow, lane occupancy and their changes over recent time windows. A prediction model estimates near-future traffic states for each segment. A control and optimization layer converts these predictions into speed recommendations, subject to safety constraints such as a maximum change between neighbouring segments and between consecutive time steps. The recommendations and the predicted congestion risk are shown on a dashboard.

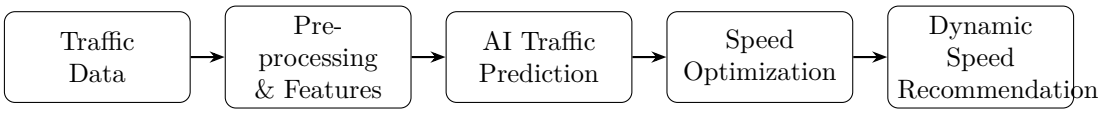


Figure 1: Pipeline of the proposed dynamic speed management system.

6 Candidate Models and Methods

Based on the literature above, we plan to benchmark the following methods.

- **Statistical baselines:** Historical average and ARIMA/SARIMA, which remain strong for short horizons and are required for a fair comparison.
- **Gradient boosting (XGBoost, LightGBM):** Fast tabular models trained on lagged features; useful when data is limited.
- **LSTM / GRU:** Recurrent networks that model temporal patterns for each segment separately.
- **Spatio-temporal GNNs (STGCN, DCRNN, Graph WaveNet):** Models that capture the influence of upstream and downstream segments.
- **Rule-based VSL (SPECIALIST-style):** A shock-wave-based controller used as the main control baseline.
- **Feedback MTFC:** An occupancy-based feedback controller, simple and robust in practice.
- **MPC or MARL controller:** The optimization- or learning-based controller that uses the forecasts proactively.

7 Datasets and Simulation Tools

Forecasting models are evaluated on real data, while controllers must be evaluated in a calibrated simulator because they cannot be tested on public roads during the project. Table 1 lists the resources we identified.

Resource	Type	Use in the project
METR-LA, PEMS-BAY	Loop detector speeds, 5-min intervals	Standard forecasting benchmarks [5]
PeMSD4, PeMSD8	Loop detector flows	Flow forecasting benchmarks
highD [22]	Drone-based trajectories, 110,000 vehicles	Calibration of car-following behaviour
IBB Traffic Graph Data [23]	2,451 sensors in Istanbul, hourly, 4 years	Turkey-specific forecasting benchmark
IBB Open Data [24]	Hourly speed and vehicle counts per geohash	Exploratory analysis of local patterns
Industrial partner data	Anonymised telemetry (if available)	Main local dataset if its resolution is minute-level or finer
SUMO + TraCI	Open-source microscopic simulator	Test environment for controllers; Python interface for RL

Table 1: Datasets and simulation tools relevant to the project.

8 Deployment Challenges and Solutions

1. Driver Compliance
Challenge: Speed recommendations are only effective if drivers follow them. Sadat and Çelikoğlu showed for Istanbul's D-100 freeway that the compliance level strongly affects the benefits of VSL [25].
Solution: Model compliance explicitly in simulation (for example 50%, 75% and 100%) and report results for each level. In a connected-vehicle scenario, recommendations can be delivered in-vehicle, which increases compliance.

2. Data Resolution and Quality
Challenge: Stop-and-go waves develop within minutes, while some open datasets are aggregated hourly. Sensors also produce missing or noisy values.
Solution: Use minute-level data from the industrial partner where possible, apply imputation and outlier filtering, and use hourly data only for long-term patterns.

3. Correct Detection of the Congestion Type
Challenge: The SPECIALIST field test showed that activations for the wrong type of jam reduce effectiveness [14].
Solution: Train the prediction layer to distinguish moving jams from stationary bottleneck queues and activate the controller only when appropriate.

4. Safety of Recommendations
Challenge: Large speed differences between neighbouring segments may create new disturbances or safety risks.
Solution: Constrain the recommended speeds with maximum step sizes in space and time, and apply discrete values (for example in 10 km/h steps) that drivers can follow easily.

5. Sim-to-Real Gap
Challenge: Learning-based controllers trained in simulation may behave unexpectedly on real roads.
Solution: Calibrate the simulator with real data, keep a rule-based fallback, and follow the lessons reported in field deployments [17].

9 Top Papers in Speed Harmonization and Traffic Forecasting

The following papers form the core of our project's theoretical and technical background.

1. "Traffic jams without bottlenecks: experimental evidence for the physical mechanism of the formation of a jam"
Authors: Yuki Sugiyama, Minoru Fukui, Macoto Kikuchi, et al.
Summary: This experiment on a circular track shows that jams can emerge without any bottleneck when density exceeds a critical value, and that the resulting jam cluster propagates backwards like a wave on a highway. [1]

2. "Review of Variable Speed Limits and Advisories: Theory, Algorithms, and Practice"
Authors: Xiao-Yun Lu, Steven E. Shladover
Summary: The review classifies VSL studies into simulation-based algorithm development, field implementations and combined VSL–ramp metering schemes, and compares results from Europe and the United States. [2]

3. "Dynamic Speed Limit Control to Resolve Shock Waves on Freeways: Field Test Results of the SPECIALIST Algorithm"
Authors: Andreas Hegyi, Serge P. Hoogendoorn
Summary: The paper reports a real-world test of a shock-wave-based VSL algorithm on the Dutch A12 freeway and evaluates both its traffic performance and the correctness of its activations. [14]

4. "Differential Variable Speed Limits Control for Freeway Recurrent Bottlenecks via Deep Actor-Critic Algorithm"
Authors: Yuankai Wu, Huachun Tan, Bin Ran
Summary: The authors propose a deep reinforcement learning controller that sets different, dynamic speed limits for each lane and show improvements in efficiency, safety and emissions in simulation. [6]

5. "Spatial-Temporal Traffic Flow Control on Motorways Using Distributed Multi-Agent Reinforcement Learning"
Authors: Krešimir Kušić, Edouard Ivanjko, Filip Vrbanić, Martin Gregurić, Ivana Dusparic
Summary: Cooperating RL agents control four motorway segments and adjust not only the speed limits but also the position and length of the VSL zones, outperforming static-zone baselines in SUMO. [7]

6. "Short-term traffic forecasting: Where we are and where we're going"
Authors: Eleni I. Vlahogianni, Matthew G. Karlaftis, John C. Golias
Summary: The paper reviews three decades of short-term forecasting research and identifies ten challenges for the next generation of forecasting models. [4]

7. "Graph Neural Network for Traffic Forecasting: A Survey"
Authors: Weiwei Jiang, Jiayun Luo
Summary: A comprehensive survey of GNN-based traffic forecasting that also collects open datasets and source code. [5]

8. "A Review of Stop-and-Go Traffic Wave Suppression Strategies: Variable Speed Limit vs. Jam-Absorption Driving"
Authors: Zhengbing He, Jorge Laval, Yu Han, Andreas Hegyi, Ryosuke Nishi, Cathy Wu
Summary: The review compares infrastructure-based VSL with vehicle-based jam-absorption driving. VSL has a stronger theoretical basis and a field test, but requires costly infrastructure and has largely overlooked secondary waves. [27]

10 Similar Systems

- **Dutch A12 dynamic speed limits:** Field trials on the Dutch network lowered speed limits during congestion and heavy rain. The evaluation reported improved throughput and safety, good user acceptance and very limited side effects [28].
- **UK M25 variable speed control:** One of the earliest European deployments, where speed limits were varied in response to traffic demand on parts of the M25 around London [29].
- **CIRCLES project, I-24 Nashville:** In November 2022, 100 vehicles with a custom adaptive cruise controller were deployed on I-24. The experiment showed that automated vehicles making up about 5% of traffic can smooth stop-and-go waves and reduce fuel consumption for all vehicles [30, 31].
- **MARL-based VSL on I-24:** A field deployment of multi-agent reinforcement learning VSL controllers on a US interstate [17].
- **Istanbul D-100 VSL studies:** Simulation studies on a 5.2 km section of the D-100, calibrated with Istanbul Metropolitan Municipality sensor data, showed reductions in total travel time and occupancy, and found occupancy-based feedback control to outperform a threshold-based decision tree [25, 26].

11 Conclusion

The literature shows that stop-and-go waves arise from unstable flow at high density, that VSL is an established way to suppress them, and that the safety benefits of VSL are clearer than its throughput benefits. Rule-based and feedback controllers such as SPECIALIST and MTFC are proven in the field and serve as strong baselines. Reinforcement learning, and especially multi-agent RL, has recently moved from simulation to field deployment. In parallel, spatio-temporal graph neural networks have become the state of the art for short-term traffic forecasting.

These two research lines are, however, usually studied separately: forecasting papers rarely close the loop with control, and most VSL controllers react to the current state rather than to a learned forecast. In addition, the literature focuses mainly on freeways, while signal-free urban arterials and Turkish traffic conditions are covered by only a few studies. Our project addresses this gap by combining a learned short-term congestion forecast with proactive, segment-level speed recommendations, evaluated in a calibrated simulation against established baselines, with an architecture that can later integrate connected-vehicle and V2X data.

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